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Open-Economy Macroeconomics: Basic Concepts

Principles: Chapter 18

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Basic Concepts

- Closed economy
  - Economy that does not interact with other economies in the world
- Open economy
  - Economy that interacts freely with other economies around the world
  - It has imports and exports

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International Flow of Goods

- Trade surplus
  - Excess of exports over imports
  - Net exports > 0
- Trade deficit
  - Excess of imports over exports
  - Net exports < 0
- Balanced trade
  - Exports equal imports

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International Flow of Goods

- Factors that affect a country's net exports:
  - Prices of goods at home and abroad
  - Exchange rates
  - Incomes of consumers at home and abroad
  - Transportation cost
  - Government trade policies

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International Flow of Capital

- Capital Flows
  - Purchase of one country's assets residents of another country
    - physical capital investment + financial investment (stock/bonds)
- Net Capital Flows
  - Amount of foreign assets bought by domestic residents minus the amount of domestic assets bought by foreigners

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Trade Deficit or Surplus?

- What is more desirable, a trade deficit or a trade surplus?
- What do you guess is more desirable: Net capital outflow from the country or Net capital inflow to the country?

International Flow of Capital

- Flow of two types: Foreign Direct Investment and Foreign Portfolio Investment
- Factors that affect net capital outflow
  - Real interest rates and returns
  - *Economic and political risks of holding assets abroad*
  - Government policies that affect foreign ownership of domestic assets

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Net Exports=Net Capital Outflow

- When NX > 0 (trade surplus)
  - Selling more goods and services to foreigners than it is buying from them
  - From net sale of goods and services
    - Receives foreign currency
    - Buy foreign assets
    - Capital - flowing out of the country: NCO > 0

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Net Exports=Net Capital Outflow

- When NX < 0 (trade deficit)
  - Buying more goods and services from foreigners
    - Than it is selling to them
  - The net purchase of goods and services
    - Needs financed
    - Selling assets abroad
    - Capital - flowing into the country: NCO < 0

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Savings, Investment and Net Exports

- $Y = C + I + G + NX$
- $\Rightarrow Y - C - G = I + NX$
- $\Rightarrow (Y-C-T) + (T-G) = NX$
- => Savings = Investment + NX
- equivalently,
- Savings = Investment + Net Capital Outflow
- So money saved goes into financing domestic investment or purchasing foreign assets

Table 1

International Flows of Goods and Capital: Summary

This table shows the three possible outcomes for an open economy.

Trade Deficit	Balanced Trade	Trade Surplus
Exports < Imports	Exports = Imports	Exports > Imports
Net Exports < 0	Net Exports = 0	Net Exports > 0
$Y < C + I + G$	$Y = C + I + G$	$Y > C + I + G$
Saving < Investment	Saving = Investment	Saving > Investment
Net Capital Outflow < 0	Net Capital Outflow = 0	Net Capital Outflow > 0

Exchange Rates

- Nominal exchange rate
  - Rate at which a person can trade currency of one country for currency of another
- Appreciation
  - Increase in the value of a currency
    - Buy more foreign currency
- Depreciation
  - Decrease in the value of a currency
    - Buy less foreign currency

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Exchange Rates

- Depreciation in the U.S. exchange rate
  - U.S. goods - cheaper relative to foreign goods
  - Consumers at home and abroad - buy more U.S. goods and fewer goods from other countries
    - Higher exports
    - Lower imports
    - Higher net exports

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Exchange Rates

- Appreciation in the U.S. exchange rate
  - U.S. goods - more expensive compared to foreign goods
  - Consumers at home and abroad - buy fewer U.S. goods and more goods from other countries
    - Lower exports
    - Higher imports
    - Lower net exports

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Real Exchange Rate

- If one laptop costs 100$, and the exchange rate is 70 Rs per dollar
- So I need to spend 7000 Rs to buy a laptop from US
- Now if price of same laptop from India is 14,000 Rs
- Then I can exchange 14000/(70*100)=2 units of foreign good per unit domestic good

Real Exchange Rate

- Real Exchange Rate = 0.014*14000/100 = 2 units
- RER = (Nominal Exchange Rate (foreign in terms of domestic) x Domestic price) / Foreign price
- Fall in RER by increase in foreign price or decrease in domestic price or depreciation of Indian currency

Purchasing Power Parity

- According to *law of one price,* goods must cost same in all locations.
- If not, there is a scope for buying good cheaper from one place and sell it in another place for a profit.
- Such profit opportunities will be exploited resulting in zero profits, thus equating prices.
- This leads to theory of Purchasing power parity.

Purchasing Power Parity

- PPP states that a unit of currency must have same real value in all countries.
- If PPP holds, and a rupee can buy same amount of goods and services in India as it can in US (obv after trading it for dollars), what then is the RER?
- RER = 1= eP/P*
- Implies Nominal exchange rate e = P*/P,
- where P* is foreign price

Purchasing Power Parity

- So if US Dollar could buy 3.65 German marks and 627 Italian lira in 1970, but in 1998 it could buy 1.76 German marks and 1737 Italian lira, what caused this change?
- The above theory has one explanation
- $e = P/P^*$, since US followed a more inflationary money supply than Germany, value of its currency relative to German currency fell more

Purchasing Power Parity

- US Inflation rate over 1970-1998: 5.3% per year
- German Inflation rate: 3.5% per year
- Italy Inflation rate: 9.6% per year
- PPP may not hold in reality:
  - All goods cannot be easily traded
  - Even traded goods are not perfect substitutes